

Huaiyu Wei

Los Angeles, CA | hywei@atmos.ucla.edu | (310) 696-3887

[My website](#) | [Google Scholar](#) | [ResearchGate](#) | [GitHub](#)

Employment

University of California, Los Angeles (UCLA) Postdoctoral Researcher Mentors: Andrew L. Stewart & James C. McWilliams	Feb. 2024 – Present
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Education

The Hong Kong University of Science and Technology (HKUST) Ph.D. in Marine Environmental Science, Advisor: Yan Wang Thesis: Parameterizing mesoscale eddy fluxes across continental slopes	Aug. 2019 – Feb. 2024
Sun Yat-sen University (SYSU) B.S. in Physical Oceanography, Advisor: Zhan Hu Thesis: Laboratory study on wave dissipation by vegetation in combined current wave flow	Sep. 2015 – July 2019
Boston University Visiting student, Advisor: Xiaozhou Ruan	May 2023 – Aug. 2023
University of British Columbia Summer Program on Ocean and Atmosphere Systems	July 2017 – Aug. 2017

Research Interests

Eddy-topography interaction; Meridional overturning circulation; Eddy parameterization; Bottom boundary layer processes; Geophysical fluid dynamics (GFD); Machine learning

Preprints

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1. **Wei, H.**, Stewart, A. L., Medvedev, A., Srinivasan, K., Solodoch, A., Manucharyan, G., & Hogg, A. Deep learning unlocks satellite-based monitoring of the global ocean overturning circulation. *under review*. (Preprint available at <https://doi.org/10.22541/essoar.15004679/v1>).
 2. Medvedev, A., **Wei, H.**, Armour, K., Srinivasan, K., Solodoch, A., & Stewart, A. Basin-wide sea-surface observations reveal post-2000 emergence of AMOC weakening. *under review*. (Preprint available at <https://essopenarchive.org/doi/full/10.22541/essoar.15002298/v1>).

Publications

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1. **Wei, H.**, Stewart, A. L., McWilliams, J. C., & Capó, E. (2026). Formation of abyssal downwelling-favorable prograde flows via mesoscale eddy potential vorticity mixing. *Journal of Physical Oceanography*, 56(6), 1363-1385. doi.org/10.1175/JPO-D-25-0196.1.
 2. **Wei, H.**, Srinivasan, K., Stewart, A. L., Solodoch, A., Manucharyan, G., & Hogg, A. (2025). Full-depth reconstruction of long-term meridional overturning circulation variability from satellite-measurable quantities via machine learning. *Journal of Advances in Modeling Earth Systems*, 17(7), e2024MS004915. doi.org/10.1029/2024MS004915.
 3. Li, D., **Wei, H.**, & Ruan, X. (2025). The importance of eddy stirring in wind-driven coastal upwelling. *Journal of Physical Oceanography*, 55(1), 29-42. doi.org/10.1175/JPO-D-24-0066.1.
 4. **Wei, H.**, Wang, Y., & Mak, J. (2024). Parameterizing eddy buoyancy fluxes across prograde shelf/slope fronts using a slope-aware GEOMETRIC closure. *Journal of Physical Oceanography*, 54(2), 359-377. doi.org/10.1175/JPO-D-23-0152.1.
 5. Xie, C. *, **Wei, H.** *, & Wang, Y. (2023). Bathymetry-aware mesoscale eddy parameterizations across upwelling slope fronts: A machine learning-augmented approach. *Journal of Physical Oceanography*, 53(12), 2861-2891. doi.org/10.1175/JPO-D-23-0017.1. (*Contributed equally)

6. Xie, C., **Wei, H.**, & Wang, Y. (2023). Impact of parameterized isopycnal diffusivity on shelf-ocean exchanges under upwelling-favorable winds: offline tracer simulations augmented by artificial neural network. *Journal of Advances in Modeling Earth Systems*, 15(4), e2022MS003424. doi.org/10.1029/2022MS003424.
7. **Wei, H.**, Wang, Y., Stewart, A. L., & Mak, J. (2022). Scalings for eddy buoyancy fluxes across prograde shelf/slope fronts. *Journal of Advances in Modeling Earth Systems*, 14(12), e2022MS003229. doi.org/10.1029/2022MS003229. (Issue cover)
8. Hu, Z., Lian, S., Zitman, T., Wang, H., He, Z., **Wei, H.**, et al. (2022). Wave breaking induced by opposing currents in submerged vegetation canopies. *Water Resources Research*, 58(4), e2021WR031121. doi.org/10.1029/2021WR031121.
9. **Wei, H.**, & Wang, Y. (2021). Full-depth scalings for isopycnal eddy mixing across continental slopes under upwelling-favorable winds. *Journal of Advances in Modeling Earth Systems*, 13(6), e2021MS002498. doi.org/10.1029/2021MS002498. (Issue cover)
10. Hu, Z., Lian, S., **Wei, H.**, Li, Y., Stive, M., & Suzuki, T. (2021). Laboratory data on wave propagation through vegetation with following and opposing currents. *Earth System Science Data*, 13(10), 4987-4999. doi.org/10.5194/essd-13-4987-2021.

Research Support & Proposals

“Mesoscale eddy-mediated diapycnal upwelling in the ocean’s sloping bottom boundary layers” NSF proposal (Physical Oceanography Program)	Pending
PI: Huaiyu Wei, Co-PIs: Andrew Stewart, Jim McWilliams, Henri Drake.	
“Leveraging machine learning and satellite measurements to predict ocean meridional overturning circulation” NSF ACCESS Explore project (EES240055).	2024-2026
Postgraduate Studentship, HKUST	2019-2024

Talks & Seminars

1. “Formation of abyssal prograde flows via eddy potential vorticity mixing”, UC Irvine, 29th July 2025.
2. “Reconstructing meridional overturning circulation from satellite measurements via neural networks”, Boston University (20th Dec. 2024), MIT (3rd Jan. 2025), UCLA AOS department Seminar (19th Feb. 2025), and SNAP seminar (24th Feb. 2025).
3. “Parameterizing eddy mixing across continental slopes under upwelling-favorable winds”, Marine Center Spring Meeting, UCLA, 9th May 2024.
4. “Parameterizing isopycnal eddy mixing across continental slopes”, AOS Ocean Seminar, UCLA, 9th Apr. 2024.

Conference Presentations & Workshops

Ocean Sciences Meeting 2026 , Glasgow, Scotland, UK	Feb. 2026
Oral presentation: “Mesoscale eddy potential vorticity mixing drives abyssal prograde along-isobath circulation”	
NCAR Regional Ocean Modeling Workshop , Boulder, CO, US.	Oct. 2025
Regional ocean modeling with MOM6 in the Community Earth System Model framework	
California GFD Meeting 2025 , UCSB, CA, US.	Sept. 2025
Organizing Committee & Oral presentation: “Eddy potential vorticity mixing drives abyssal prograde along-isobath circulation”	
Climate Process Team Meeting 2025 , NYU/online, US.	Aug. 2025
Oral presentation: “Formation of abyssal prograde flows via mesoscale eddy potential vorticity mixing”	
AGU 2024 , Washington, DC, US.	Dec. 2024
Oral presentation: “Reconstructing meridional overturning circulation from satellite measurements via Neural Networks”	
Ocean Sciences Meeting 2024 , New Orleans, US	Feb. 2024
eLightning presentation: “Parameterizing eddy buoyancy fluxes across prograde shelf/slope fronts	

using a slope-aware GEOMETRIC closure”

Ocean Transport and Eddy Energy Meeting 2023, WHOI, US

May 2023

Oral presentation: “Parameterization for eddy buoyancy fluxes across prograde shelf/slope fronts”

Ocean Sciences Meeting 2022, Online

Mar. 2022

Oral presentation: “Full-depth scalings for isopycnal eddy mixing across continental slopes under upwelling-favorable winds”

EGU 2018, Vienna, Austria

Apr. 2018

Poster presentation: “The pattern and control of erodibility of cohesive sediments in a *Spartina alterniflora* marsh on the coast of Jiangsu, China”

Selected Awards

Richard P. and Linda S. Turco Exceptional Research Publication Award, UCLA

2026

Chancellor’s Award for Postdoctoral Research nominee, UCLA

2025

RedBird Academic Excellence Award, HKUST

2023

Best Presentation Award, HKUST Postgraduate Seminar

2023

Outstanding Graduate, SYSU

2019

National Scholarship, China (Top 1%)

2018

The Giordano Scholarship, SYSU (Top 2%)

2017

The Coca-Cola Scholarship for Outstanding Students, SYSU (Top 5%)

2016

Teaching Experience

Guest lecturer in graduate GFD at UCLA - Baroclinic instability

2025

Guest lecturer in undergraduate Physical Oceanography at UCLA - Density

2025

Guest lecturer in undergraduate Physical Oceanography at UCLA - Tides

2024

Teaching assistant at HKUST

2020-2023

Additional Information

Year of birth: 1997.

Language Skills: Mandarin (Native), English (IELTS score: 7.5).

Computer programming: Fortran, MATLAB, Python.

Journal reviewer: Communications Earth & Environment, Geophysical Research Letters, Journal of Advances in Modeling Earth Systems, Journal of Geophysical Research: Oceans, Journal of Physical Oceanography, Nature Communications, Ocean Modelling.